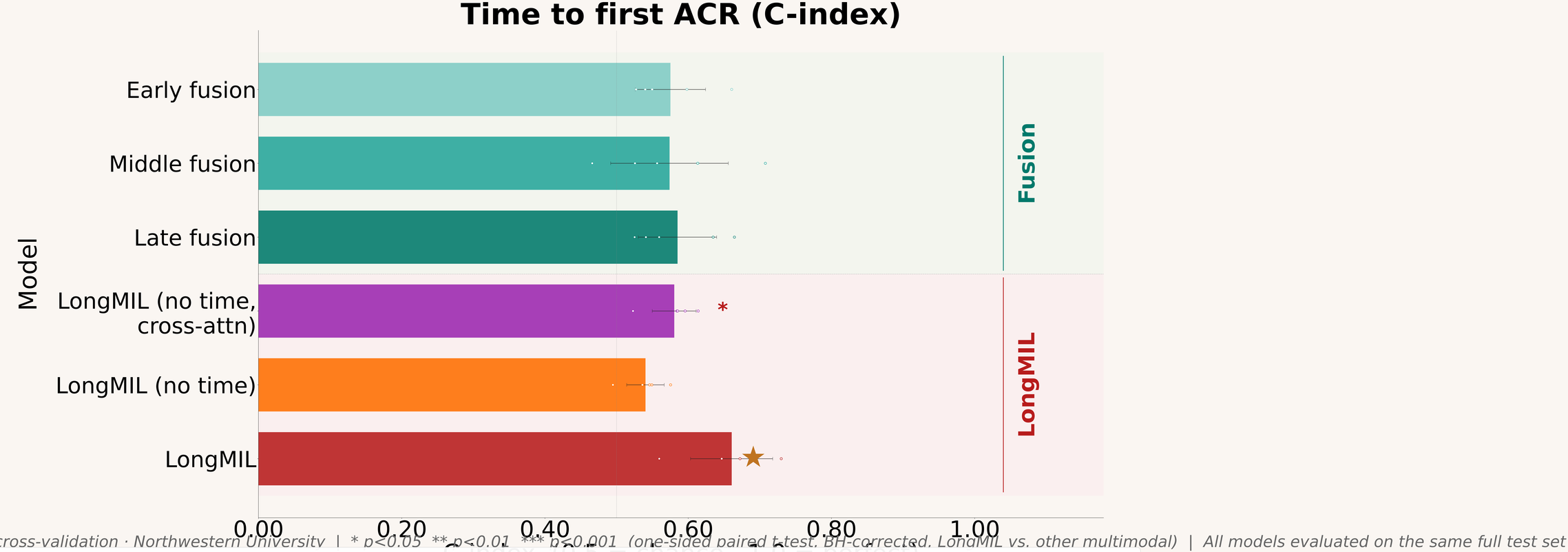
Time to first ACR (C-index)
Model
Early fusion
Middle fusion
Late fusion
LongMIL (no time, cross-attn)
LongMIL (no time)
LongMIL
Fusion
LongMIL
*
0.00
0.20
0.40
0.60
0.80
1.00
C-index (0.5 = chance, 1.0 = perfect)
n = 263 patients · 5-fold cross-validation · Northwestern University | * p<0.05 ** p<0.01 *** p≤0.001 (one-sided paired t-test, BH-corrected, LongMIL vs. other multimodal) | All models evaluated on the same full test set
Early fusion
Middle fusion
Late fusion
LongMIL (no time, cross-attn)
LongMIL (no time)
LongMIL
Chance (0.5)